

Extension: Perpendiculars and distance



1 $d = \frac{|Ax_1 + By_1 + C|}{\sqrt{A^2 + B^2}}$

2 $\frac{5}{\sqrt{65}}$

3
a $y = 7$
b $\frac{9}{\sqrt{113}}$

4 $k = 7, k = -27$

5 $k = 5, k = 35$

6
a $\frac{28}{5}$ units
b $\frac{72}{25}$ units
c $\frac{23}{13}$ units
d $2\sqrt{13}$ units

7
a $3x - 4y + 13 = 0$
b 3 units
c 15 square units

8
a Distance $= \sqrt{5}$ units
b $b = \pm\sqrt{319}$